Manage storage aggregate plexes

ONTAP 9.11.1 REST API reference

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Manage storage aggregate plexes

Storage aggregates aggregate.uuid plexes endpoint overview

Overview

The Storage Aggregate Plex API provides relevant state information for each plex in the aggregate. For each plex, details are provided for the RAID groups in the plex and the disks that make up each RAID group.

Examples

Retrieving all aggregates and plexes

The following example shows the response with a list of aggregates and plexes:

The call:

curl -X GET "https://<mgmt-ip>/api/storage/aggregates/*/plexes" -H "accept: application/json"</mgmt-ip>

The response:

{ "records": [{ "aggregate": { "uuid": "04b7296e-a302-42a6-a2a9-dda6be054b29", "name": "test2" }, "name": "plex0" }, { "aggregate": { "uuid": "04b7296e-a302-42a6-a2a9-dda6be054b29", "name": "test2" }, "name": "plex1", }, { "aggregate": { "uuid": "66c4b221-65ff-4211-9b58-ada3c6fc41af", "name": "test" }, "name": "plex0" }, { "aggregate": { "uuid": "66c4b221-65ff-4211-9b58-ada3c6fc41af", "name": "test" }, "name": "plex1" }, { "aggregate": { "uuid": "7ee89e48-5d81-4609-9e1b-5d8d0995a886", "name": "aggr1", }, "name": "plex0", }, { "aggregate": { "uuid": "8bb2e3bf-c4f1-4748-9033-ca9231cf1c40", "name": "test3", }, "name": "plex0", }, { "aggregate": { "uuid": "8bb2e3bf-c4f1-4748-9033-ca9231cf1c40", "name": "test3", }, "name": "plex1" }, { "aggregate": { "uuid": "8f13de5c-99cf-4ada-884c-3cc32deb304a", "name": "aggr2", }, "name": "plex0" }], "num_records": 8 }

```
### Retrieving a specific plex in all aggregates
The following example shows the response with a list of specific plexes in
all aggregates:


# The call:
curl -X GET "https://<mgmt-ip>/api/storage/aggregates/*/plexes/plex0" -H
"accept: application/json"

# The response:
{
"records": [
  {
    "aggregate": {
      "uuid": "04b7296e-a302-42a6-a2a9-dda6be054b29",
```

```
      "name": "test2"
    },
    "name": "plex0"
  },
  {
    "aggregate": {
      "uuid": "66c4b221-65ff-4211-9b58-ada3c6fc41af",
      "name": "test"
    },
    "name": "plex0"
  },
  {
    "aggregate": {
      "uuid": "7ee89e48-5d81-4609-9e1b-5d8d0995a886",
      "name": "aggr1",
    },
    "name": "plex0",
  },
  {
    "aggregate": {
      "uuid": "8bb2e3bf-c4f1-4748-9033-ca9231cf1c40",
      "name": "test3",
    },
    "name": "plex0",
  },
  {
    "aggregate": {
      "uuid": "8f13de5c-99cf-4ada-884c-3cc32deb304a",
      "name": "aggr2",
    },
    "name": "plex0"
  }
],
"num_records": 8
}
```


Retrieving the list of plexes in an aggregate

The following example shows the response with the list of plexes in an aggregate:

```
# The API:
/api/storage/aggregates/{uuid}/plexes

# The call:
curl -X GET "https://<mgmt-ip>/api/storage/aggregates/19425837-f2fa-4a9f-
8f01-712f626c983c/plexes" -H "accept: application/json"

# The response:
{
"records": [
  {
    "name": "plex0",
  },
  {
    "name": "plex4",
  }
],
"num_records": 2,
}
```

Retrieving a specific plex in an aggregate

The following example shows the response when requesting a specific plex of an aggregate:

```
# The API:
/api/storage/aggregates/{uuid}/plexes/{name}

# The call:
curl -X GET "https://<mgmt-ip>/api/storage/aggregates/19425837-f2fa-4a9f-
8f01-712f626c983c/plexes/plex0" -H "accept: application/json"

# The response:
{
"aggregate": {
  "uuid": "19425837-f2fa-4a9f-8f01-712f626c983c",
  "name": "test1",
},
"name": "plex0",
"online": true,
"state": "normal",
"pool": "pool0",
"resync": {
  "active": false
},
"raid_groups": [
```

```
  {
    "name": "rg0",
    "raid_type": "raid_dp",
    "cache_tier": false,
    "degraded": false,
    "recomputing_parity": {
      "active": false
    },
    "reconstruct": {
      "active": false
    },
    "disks": [
      {
        "position": "dparity",
        "state": "normal",
        "type": "ssd",
        "usable_size": 86769664,
        "disk": {
          "name": "1.1.29",
        }
      },
      {
        "position": "parity",
        "state": "normal",
        "type": "ssd",
        "usable_size": 86769664,
        "disk": {
          "name": "1.1.4",
        }
      },
      {
        "position": "data",
        "state": "normal",
        "type": "ssd",
        "usable_size": 86769664,
        "disk": {
          "name": "1.1.30",
        }
      },
      {
        "position": "data",
        "state": "normal",
        "type": "ssd",
        "usable_size": 86769664,
        "disk": {
          "name": "1.1.5",
```

```
          }
        },
        {
          "position": "data",
          "state": "normal",
          "type": "ssd",
          "usable_size": 86769664,
          "disk": {
            "name": "1.1.31",
          }
        },
        {
          "position": "data",
          "state": "normal",
          "type": "ssd",
          "usable_size": 86769664,
          "disk": {
            "name": "1.1.6",
          }
        }
      ]
    }
  ],
  }
```


Retrieve a collection of plexes for an aggregate

GET `/storage/aggregates/{aggregate.uuid}/plexes`

Introduced In: 9.6

Retrieves the collection of plexes for the specified aggregate.

Related ONTAP commands

- `storage aggregate plex show`

Parameters

Name	Type	In	Required	Description
aggregate.uuid	string	path	True	Aggregate UUID
aggregate.name	string	query	False	Filter by aggregate.name

Name	Type	In	Required	Description
state	string	query	False	Filter by state
raid_groups.reconstruct.active	boolean	query	False	Filter by raid_groups.reconstruct.active • Introduced in: 9.7
raid_groups.reconstruct.percent	integer	query	False	Filter by raid_groups.reconstruct.percent • Introduced in: 9.7
raid_groups.raid_type	string	query	False	Filter by raid_groups.raid_type • Introduced in: 9.9
raid_groups.recomputing_parity.active	boolean	query	False	Filter by raid_groups.recomputing_parity.active • Introduced in: 9.7
raid_groups.recomputing_parity.percent	integer	query	False	Filter by raid_groups.recomputing_parity.percent • Introduced in: 9.7
raid_groups.cache_tier	boolean	query	False	Filter by raid_groups.cache_tier • Introduced in: 9.7

Name	Type	In	Required	Description
raid_groups.degraded	boolean	query	False	Filter by raid_groups.degraded • Introduced in: 9.7
raid_groups.disks.disk.name	string	query	False	Filter by raid_groups.disks.disk.name • Introduced in: 9.7
raid_groups.disks.position	string	query	False	Filter by raid_groups.disks.position • Introduced in: 9.7
raid_groups.disks.usable_size	integer	query	False	Filter by raid_groups.disks.usable_size • Introduced in: 9.7
raid_groups.disks.state	string	query	False	Filter by raid_groups.disks.state • Introduced in: 9.7
raid_groups.disks.type	string	query	False	Filter by raid_groups.disks.type • Introduced in: 9.7
raid_groups.name	string	query	False	Filter by raid_groups.name • Introduced in: 9.7
pool	string	query	False	Filter by pool

Name	Type	In	Required	Description
name	string	query	False	Filter by name
online	boolean	query	False	Filter by online
resync.active	boolean	query	False	Filter by resync.active
resync.percent	integer	query	False	Filter by resync.percent
resync.level	string	query	False	Filter by resync.level
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
error	error	
num_records	integer	Number of records
records	array[plex]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  },
  "records": {
    "aggregate": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "aggr1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "name": "plex0",
    "pool": "pool0",
    "raid_groups": {
      "disks": {
        "disk": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "1.0.1"
        },
        "position": "data",
        "state": "normal",
        "type": "ssd",
        "usable_size": 947912704
```

```
      },
      "name": "rg0",
      "raid_type": "raid_dp",
      "recomputing_parity": {
        "percent": 10
      },
      "reconstruct": {
        "percent": 10
      }
    },
    "resync": {
      "level": "full",
      "percent": 10
    },
    "state": "normal"
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

_links

Name	Type	Description
self	href	

aggregate

Name	Type	Description
_links	_links	
name	string	
uuid	string	

disk

Name	Type	Description
_links	_links	
name	string	

raid_group_disk

Name	Type	Description
disk	disk	
position	string	The position of the disk within the RAID group.
state	string	The state of the disk within the RAID group.
type	string	Disk interface type
usable_size	integer	Size in bytes that is usable by the aggregate.

recomputing_parity

Name	Type	Description
active	boolean	RAID group is recomputing parity
percent	integer	Recomputing parity percentage

reconstruct

Name	Type	Description
active	boolean	One or more disks in this RAID group are being reconstructed.
percent	integer	Reconstruct percentage

raid_group

Name	Type	Description
cache_tier	boolean	RAID group is a cache tier

Name	Type	Description
degraded	boolean	RAID group is degraded. A RAID group is degraded when at least one disk from that group has failed or is offline.
disks	array[raid_group_disk]	
name	string	RAID group name
raid_type	string	RAID type of the raid group.
recomputing_parity	recomputing_parity	
reconstruct	reconstruct	

resync

Name	Type	Description
active	boolean	Plex is being resynchronized to its mirrored plex
level	string	Plex resyncing level
percent	integer	Plex resyncing percentage

plex

Name	Type	Description
aggregate	aggregate	
name	string	Plex name
online	boolean	Plex is online
pool	string	SyncMirror pool assignment
raid_groups	array[raid_group]	
resync	resync	
state	string	Plex state

Retrieve a plex specified by the aggregate UUID and plex name

GET `/storage/aggregates/{aggregate.uuid}/plexes/{name}`

Introduced In: 9.6

Retrieves the plex specified by the aggregate UUID and plex name.

Related ONTAP commands

- `storage aggregate plex show`

Parameters

Name	Type	In	Required	Description
aggregate.uuid	string	path	True	Aggregate UUID
name	string	path	True	Plex name
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
aggregate	aggregate	
name	string	Plex name
online	boolean	Plex is online
pool	string	SyncMirror pool assignment
raid_groups	array[raid_group]	
resync	resync	
state	string	Plex state

Example response


```
{
  "aggregate": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "aggr1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "plex0",
  "pool": "pool0",
  "raid_groups": {
    "disks": {
      "disk": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "1.0.1"
      },
      "position": "data",
      "state": "normal",
      "type": "ssd",
      "usable_size": 947912704
    },
    "name": "rg0",
    "raid_type": "raid_dp",
    "recomputing_parity": {
      "percent": 10
    },
    "reconstruct": {
      "percent": 10
    }
  },
  "resync": {
    "level": "full",
    "percent": 10
  },
  "state": "normal"
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

aggregate

Name	Type	Description
_links	_links	
name	string	
uuid	string	

disk

Name	Type	Description
_links	_links	
name	string	

raid_group_disk

Name	Type	Description
disk	disk	
position	string	The position of the disk within the RAID group.
state	string	The state of the disk within the RAID group.
type	string	Disk interface type
usable_size	integer	Size in bytes that is usable by the aggregate.

recomputing_parity

Name	Type	Description
active	boolean	RAID group is recomputing parity
percent	integer	Recomputing parity percentage

reconstruct

Name	Type	Description
active	boolean	One or more disks in this RAID group are being reconstructed.
percent	integer	Reconstruct percentage

raid_group

Name	Type	Description
cache_tier	boolean	RAID group is a cache tier
degraded	boolean	RAID group is degraded. A RAID group is degraded when at least one disk from that group has failed or is offline.
disks	array[raid_group_disk]	
name	string	RAID group name
raid_type	string	RAID type of the raid group.
recomputing_parity	recomputing_parity	
reconstruct	reconstruct	

resync

Name	Type	Description
active	boolean	Plex is being resynchronized to its mirrored plex
level	string	Plex resyncing level
percent	integer	Plex resyncing percentage

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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